

Comparing indices of diet quality with chronic disease mortality risk in postmenopausal women in the Women's Health Initiative Observational Study:...

Poor diet quality is thought to be a leading risk factor for years of life lost. We examined how scores on 4 commonly used diet quality indices-the Healthy Eating Index 2010 (HEI), the Alternative Healthy Eating Index 2010 (AHEI), the Alternate Mediterranean Diet (aMED), and the Dietary Approaches to Stop Hypertension (DASH)-are related to the risks of death from all causes, cardiovascular disease (CVD), and cancer among postmenopausal women. Our prospective cohort study included 63,805 participants in the Women's Health Initiative Observational Study (from 1993-2010) who completed a food frequency questionnaire at enrollment. Cox proportional hazards models were fit using person-years as the underlying time metric. We estimated multivariate hazard ratios and 95% confidence intervals for death associated with increasing quintiles of diet quality index scores. During 12.9 years of follow-up, 5,692 deaths occurred, including 1,483 from CVD and 2,384 from cancer. Across indices and after adjustment for multiple covariates, having better diet quality (as assessed by HEI, AHEI, aMED, and DASH scores) was associated with statistically significant 18%-26% lower all-cause and CVD mortality risk. Higher HEI, aMED, and DASH (but not AHEI) scores were associated with a statistically significant 20%-23% lower risk of cancer death. These results suggest that postmenopausal women consuming a diet in line with a priori diet quality indices have a lower risk of death from chronic disease. Published by Oxford University Press on behalf of the Johns Hopkins Bloomberg School of Public Health 2014. This work is written by (a) US Government employee(s) and is in the public domain in the US.